Lecture 3 Homework

Programming

1. Write and test code for the Fast Exponentiation Algorithm.

Exercises

1. Find the multiplicative inverse of $7 \bmod 59$ by computing $7^{57} \bmod 59$. Explain why it works.
2. Compute $2^{56}\%1001$ using the Fast Exponentiation Algorithm.

Suggested Reading

- Garrett – Sections 7.8, 10.3, 10.4, 12.5
- Smart – Sections 1.2, 7.4, 10.1, 11.2, 15.2.2, 17.3.1, A.8
- Forouzan – 4.1, 9.6, 10.4, 15.3, 15.4